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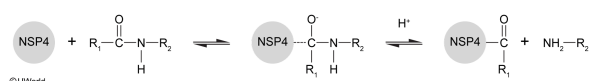


Figure 1 Degradation of viral proteins by acylation of NSP4

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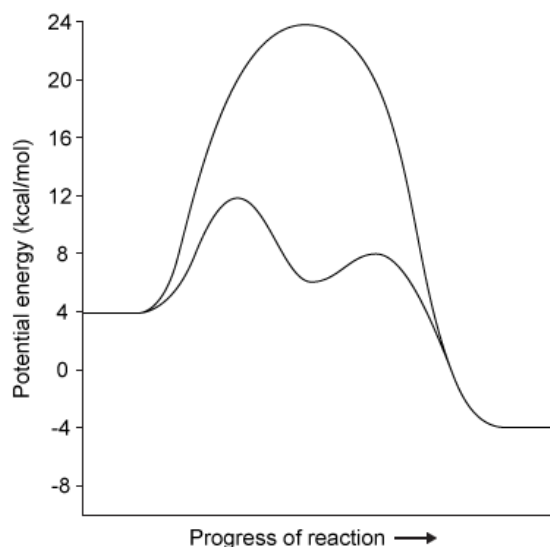


Figure 2 Free energy profiles of synthetic viral protein cleavage

Knowledge of amino acid frequency distribution at an enzymatic active site may facilitate the design of novel active sites and enzyme-specific inhibitors. Researchers compared NSP4 sequences from various species and analyzed the composition of amino acids that form the NSP4 active site during protein folding to study the characteristics of residues involved in active site formation (Figure 3).

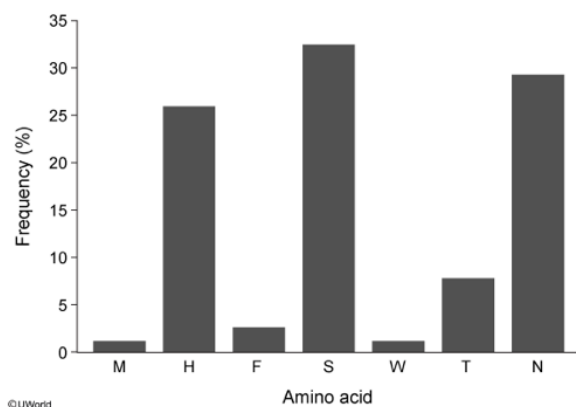


Figure 3 Frequency distribution of amino acid residues at the NSP4 active site

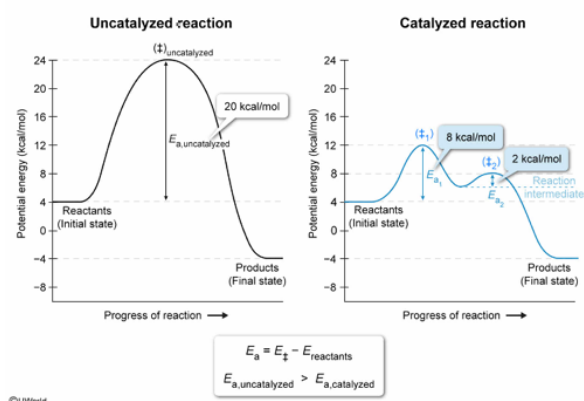
What is the activation energy for the uncatalyzed reaction in Figure 2?

- ☐ A. 12 kcal/mol [4%]
- ✓ ☒ B. 20 kcal/mol [77%]
- ☐ C. 24 kcal/mol [17%]
- ☐ D. 28 kcal/mol

Correct

77% answered correctly

Explanation:



In a free energy diagram, the x-axis represents the progress of a reaction (reaction coordinate) and the y-axis depicts the Gibbs free energy G of the system. Reactants and products are separated by an energy barrier known as the **energy of activation E_a** , defined as the **energy difference between the transition state ($\ddagger$) and the reactants** in a reaction (ie, $E_a = E_{\ddagger} - E_{\text{reactants}}$). The transition state of a reaction is the intermediate point of **greatest energy** where bonds break and form as the product is generated. For a reaction to proceed, the reactants must possess sufficient energy to overcome E_a . The height of the E_a barrier determines the reaction rate (ie, the amount of product formed per unit time); the greater the E_a , the slower the reaction.

Uncatalyzed reactions have a greater E_a than catalyzed reactions, and therefore are slower. Without a catalyst, only a few molecules have sufficient energy to overcome the E_a barrier and achieve the transition state ($\ddagger_{\text{uncatalyzed}}$). However, the addition of a catalyst increases the rate of a reaction by lowering the E_a barrier, allowing more reactants to achieve the transition state and form the product. Figure 2 shows the reaction pathway of an uncatalyzed reaction compared to a reaction catalyzed by NSP4. Using the given formula for the uncatalyzed reaction, E_a is calculated as $24 - 4 = 20$ kcal/mol.

(Choice A) The catalyzed reaction shows that only 8 kcal/mol and 4 kcal/mol are needed to achieve the first ($\ddagger_1$) and second ($\ddagger_2$) transition states, respectively, when starting from the reactants. The sum of these energies is 12 kcal/mol.

(Choice C) The total energy of the transition state for the *uncatalyzed* reaction is 24 kcal/mol.

(Choice D) This value represents E_a of the reverse reaction, which is the energy difference between the transition state and the products (ie, $E_{a,\text{reverse}} = E_{\ddagger} - E_{\text{products}}$), and is equivalent to 28 kcal/mol.

Educational objective:

The activation energy E_a is the minimum energy required for reactants to reach the transition state and initiate a reaction (given by $E_a = E_{\ddagger} - E_{\text{reactants}}$). Uncatalyzed reactions have greater E_a than catalyzed reactions, and therefore are slower. Catalysts allow the reaction to proceed at a faster rate by lowering the activation energy.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400320
System:
Enzymes

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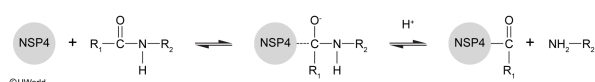


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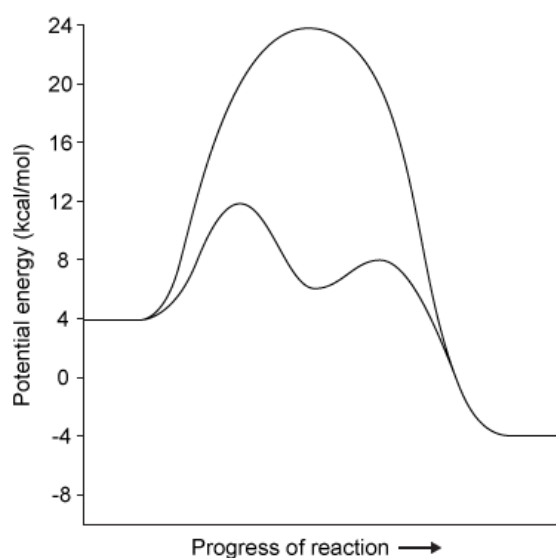


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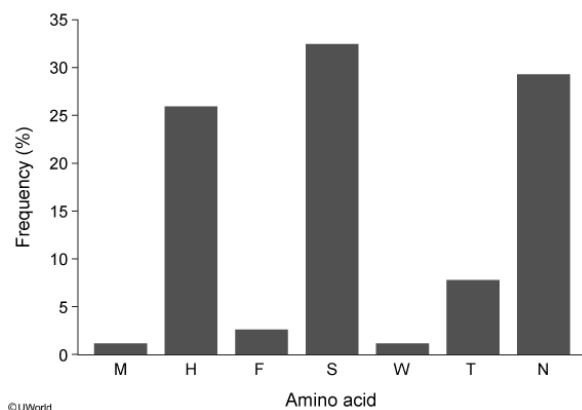
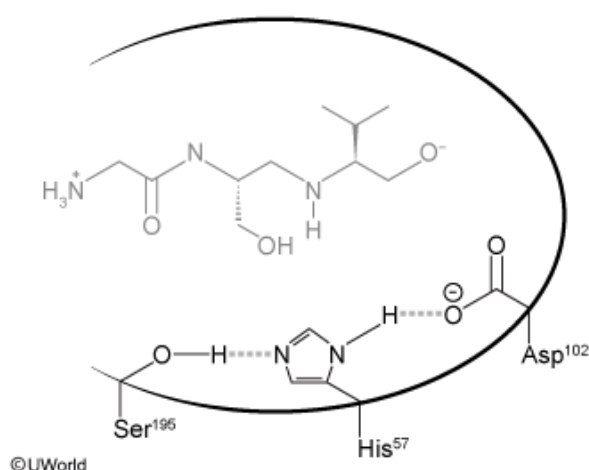


Figure 3 Frequency distribution of amino acid residues at the NSP4 active site

The figure below shows the active site of NSP4 with three residues that are essential for catalysis.



If site-directed mutagenesis was used to create a missense mutation at His⁵⁷, this would likely result in which of the following changes regarding the enzyme-catalyzed degradation of the viral peptide depicted in the active site?

- I. The time required for the reaction to achieve equilibrium would increase.
 - II. The equilibrium constant K_{eq} of the reaction would decrease.
 - III. The enzyme-substrate dissociation constant K_d would increase.
- ☐ A. I only [11%]
☐ B. III only [10%]
☐ C. I and II only [14%]
☒ D. I and III only [64%]

Correct

64% answered correctly

Explanation:

Enzyme kinetic parameters	Effects on reaction	
	Fully functional enzyme	Deleterious mutation in enzyme
Activation energy (E_a)	Decrease	Increase
Reaction rate	Increase	Decrease
Equilibrium constant (K_{eq})	No change	No change
Gibbs free energy change (ΔG)	No change	No change
Michaelis-Menten constant (K_M)	Values are enzyme-substrate specific	Increase
Dissociation constant (K_d)		Increase
Turnover rate (k_{cat})	Values are enzyme-substrate specific and dependent on [E]	Decrease
Catalytic efficiency (k_{cat}/K_M)		
Maximum velocity (V_{max})		

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Enzymes are biological catalysts that lower the activation energy of a reaction, allowing the reaction to proceed at a faster rate. Substrate molecules bind to the enzyme's active site, where catalytic activity occurs. The enzyme and substrate undergo conformational changes that make the transition state more stable and favorable, resulting in lower activation energy. **Enzyme function** is sensitive to changes in pH and temperature as well as **deleterious mutations**, particularly those involving the **active site**.

A missense mutation at the histidine in position 57 of the NSP4 sequence (ie, His⁵⁷) would result in a different amino acid at this site and alter the interaction between the viral peptide (substrate) and the NSP4 active site. This mutation would be expected to decrease the binding affinity of the enzyme for its substrate, resulting in an **increased dissociation constant K_d** for the enzyme-substrate complex (**Number III**). Even when saturated, mutations at the active site would likely interfere with normal enzymatic function and lead to a **decreased reaction rate** (ie, increasing the time the reaction takes to reach equilibrium) (**Number I**). If the mutation is on a conserved residue required for catalysis, the reaction rate may even drop to the uncatalyzed rate.

(Number II) Enzymes *do not alter* the free energy change ΔG or the **equilibrium constant K_{eq}** of a reaction. The overall ΔG depends only on the energy difference between products and reactants ($\Delta G = G_{\text{products}} - G_{\text{reactants}}$) and can be visualized in a **free energy diagram**. The equilibrium constant K_{eq} of a reaction is the ratio of products to reactants for a given reaction at equilibrium at a specific temperature and is related to ΔG (ie, $\Delta G = -RT \ln K_{eq}$). Although a functional enzyme would decrease the time taken for the reaction to reach equilibrium (ie, $\Delta G = 0$ for the reaction) by increasing the reaction rate, it would not alter K_{eq} .

Educational objective:

Enzymes do not alter the free energy change ΔG or the equilibrium constant K_{eq} of a reaction, but by increasing the reaction rate they do decrease the time taken for the reaction to reach equilibrium. Deleterious mutations involving the enzyme's active site are likely to alter the enzyme's affinity for its substrate (ie, increase K_d) and interfere with enzymatic function (ie, decrease reaction rate).

Subject:
Biochemistry

System:
Enzymes

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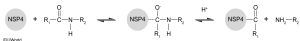


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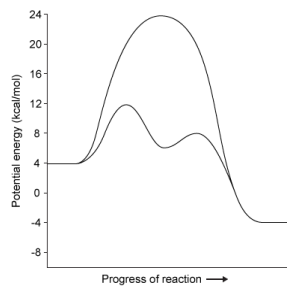


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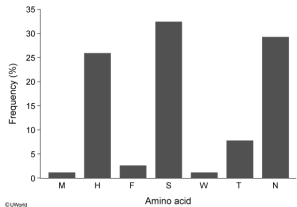


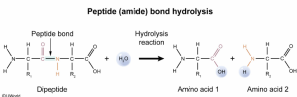
Figure 3 Frequency distribution of amino acid residues at the NSP4 active site

Considering that NSPs are proteases, which of the following statements best describes the reaction catalyzed by NSPs?

- ☐ A. Water is produced by the reaction. [13%]
- ☒ B. Water is consumed by the reaction. [71%]
- ☐ C. Water is produced and then consumed by the reaction. [6%]
- ☐ D. Water is consumed and then produced by the reaction. [7%]

Correct
71% answered correctly

Explanation:



Peptides are linear sequences of amino acids linked by amide bonds (ie, --CO--NH--) that join the α -carboxyl group of one amino acid to the α -amino group of another amino acid. Proteins that have completed their biological purpose must be degraded by **proteases** so that their constituent amino acids can be reused to generate new peptides. To cleave an amide bond during protein degradation, a water molecule (H_2O) must be added using acids (H^+) or bases (OH^-) in a reaction known as amide hydrolysis.

Figure 1 shows the acylation process (ie, R--C=O addition) of **amide hydrolysis** by NSP4, which cleaves the amide bond of the viral peptide by adding a hydrogen atom to the nitrogen. The hydrogen atom displaces the C--N single bond when electrons from the oxygen on the carbonyl carbon reform the double bond. The acyl group ($\text{R}_1\text{--C=O}$) attached to NSP4 is subsequently converted to a carboxylic acid when electrons from an OH^- group attack the electrophilic (electron-attracting) carbonyl carbon.

(Choices A, C, and D) Hydrolysis reactions involve only the consumption of water to break bonds. They do not involve water production, which occurs in condensation reactions such as **peptide bond formation**.

Educational objective:

Proteins can be degraded by protease enzymes. During peptide hydrolysis, a water molecule is used to cleave the C--N bond in the amide linkage using an acid (H^+) or base (OH^-). In contrast, the reverse of amide hydrolysis is amide bond condensation, which involves the formation of a water molecule from two amino acids to form a larger peptide.

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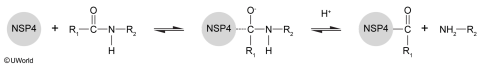


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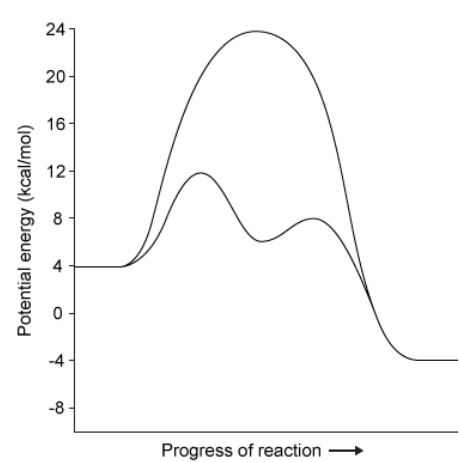


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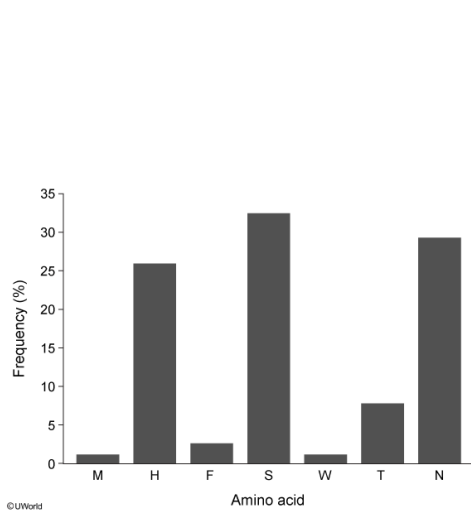


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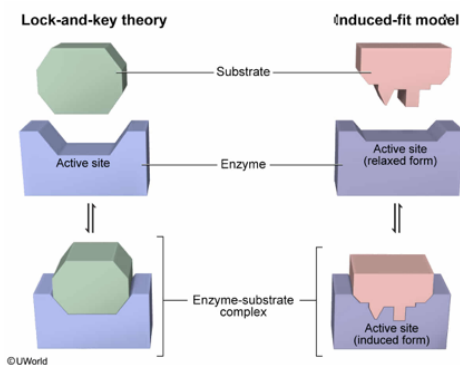
Which of the following is the most accurate statement regarding NSP4 binding of viral proteins?

- ✓ ☒ A. NSP4 undergoes an energy-requiring conformational change upon binding. [60%]
☐ B. NSP4 quaternary structure must undergo conformational changes upon binding. [18%]
☐ C. Heat is consumed when NSP4 binds its substrate. [9%]
☐ D. An NSP4 holoenzyme is formed after the release of products. [12%]

Correct

60% answered correctly

Explanation:



Two competing theories exist regarding enzyme-substrate interaction, the lock-and-key theory and the induced fit model. The **lock-and-key theory** proposes that an enzyme's active site is already in the proper structural conformation to allow a substrate to bind readily and form an active enzyme-substrate (ES) complex. This theory proposes that *no conformational changes are necessary* for catalysis to occur.

The **induced fit model** states that a **conformational change is induced** when the substrate binds the active site, resulting in the formation of a functional ES complex that is said to be in the **induced form**. Enzymes are highly specific for their substrates, and only the correct substrate can induce the required complementary change in active site conformation. Shifting an enzyme's tertiary or quaternary structure to alter the shape of the active site, combined with the corresponding conformational changes in the substrate toward the reaction's **transition state**, **requires energy**. The passage states that NSP4 alters its tertiary

structure upon substrate binding, so NSP4 most likely undergoes a conformational change that requires energy upon substrate binding.

(Choice B) Quaternary structure is only present in multimeric proteins that have multiple subunits. The passage states that NSP4 is a monomer (one subunit), so it does not have a quaternary structure.

(Choice C) When the correct substrate enters and is bound by an enzyme's active site, heat is released (*not consumed*) from the formation of weak bonds (eg, hydrogen, hydrophobic, ionic bonds) between the two molecules.

(Choice D) Cofactors are molecules required for enzyme catalysis. Enzymes that require cofactors for activity are called apoenzymes when the cofactor is absent whereas those containing cofactors are known as holoenzymes. The passage does not indicate that NSP4 needs a cofactor for activity.

Educational objective:

The lock and key model proposes that the active site of an enzyme is already in the correct conformation to bind to a substrate. In contrast, the induced fit theory states that the initial binding of a substrate causes the active site to undergo a conformational change that complements the structure of the substrate.

subject:
Biochemistry

system:
Enzymes

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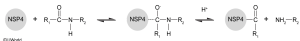


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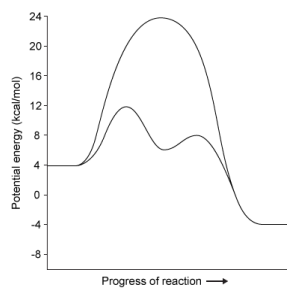


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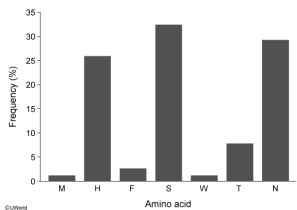


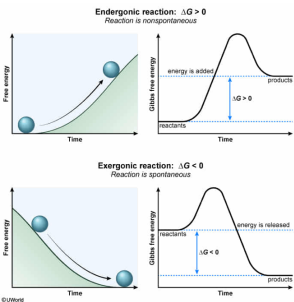
Figure 3 Frequency distribution of amino acid residues at the NSP4 active site

Based on the passage, the cleavage of viral proteins by NSP4 is:

- ☐ A. an exergonic process moving in the reverse direction. [3%]
- ☐ B. an endergonic process moving in the forward direction. [6%]
- ☐ C. a nonspontaneous reaction requiring energy to proceed. [11%]
- ☒ D. a spontaneous reaction resulting in a release of energy. [78%]

Correct
79% answered correctly

Explanation:



In thermodynamics, the spontaneity of a reaction is determined by assessing its **Gibbs free energy change**. Gibbs free energy *G* is the potential energy available in a substance that can be used for chemical work. The difference in potential energy between reactants and products represents the ΔG of a reaction. Biochemical reactions are classified based on ΔG , which indicates whether a reaction requires external energy.

Spontaneous reactions proceed without energy from external sources and have **negative ΔG** values because the reactants possess greater free energy than the products; this imbalance causes a **release of energy** characteristic of **exergonic** reactions. The energy released comes from the breaking of bonds in the reactants to form the products, which consequently have lower potential energy. Based on Figure 2, the **energy difference** between the reactants (NSP4 and viral proteins) and the products (acylated NSP4 and hydrolyzed proteins) produces a negative ΔG value, indicating that the reaction releases energy.

$$\Delta G = G_{\text{products}} - G_{\text{reactants}} = -4 - 4 = -8 \text{ kcal/mol}$$

As a result, the acylation reaction by serine proteases in Figure 1 can be described as one that is spontaneous, exergonic, and moving in the forward direction (ie, reactants → products) (**Choices A and C**).

(Choice B) Reactions with positive ΔG values require energy input as the reactants have lower free energy than the products. These types of reactions are called endergonic because they must absorb energy from external sources to proceed, and therefore are nonspontaneous. For example, if the acylation reaction shown in Figure 1

were reversed (ie, products $\rightarrow$ reactants), energy would be required to reform the viral protein.

Educational objective:

Endergonic reactions are nonspontaneous reactions that absorb energy because their Gibbs free energy change ΔG is positive. Exergonic reactions are spontaneous reactions with negative ΔG values that release energy as they proceed.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400324
System:
Enzymes

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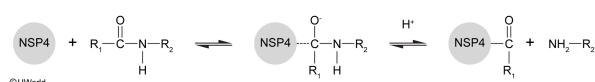


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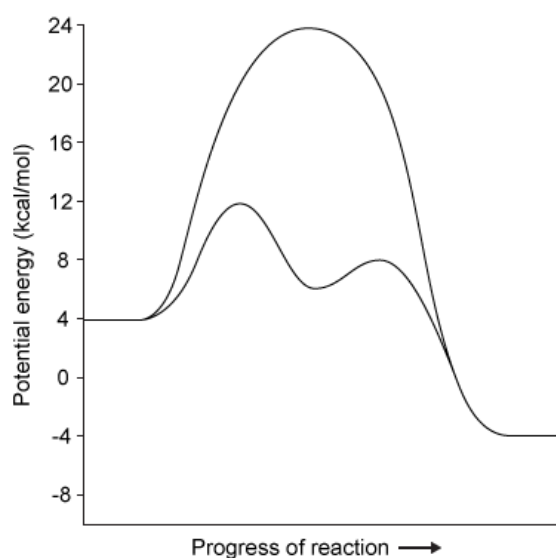


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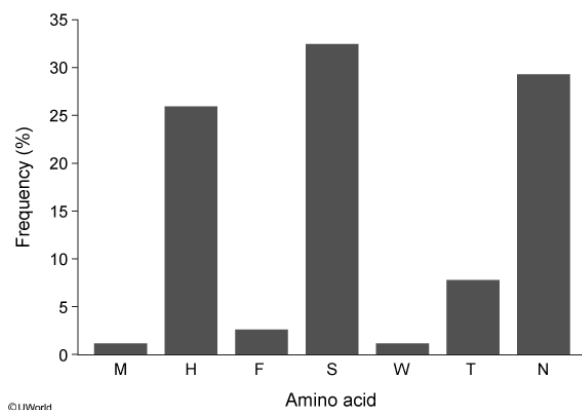


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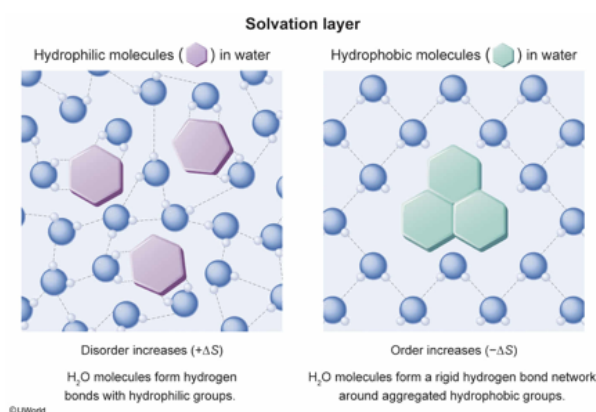
As NSP4 folds, the water molecules surrounding it become:

- ☐ A. more ordered; ΔS of water is negative. [25%]
- ☐ B. more ordered; ΔS of water is positive. [11%]
- ✓ ☒ C. more disordered; ΔS of water is positive. [56%]
- ☐ D. more disordered; ΔS of water is negative. [6%]

Correct

57% answered correctly

Explanation:



The polarity and charge of amino acid residues on the surface of a protein affect the order of the surrounding water molecules, as measured by entropy, ΔS . Increased order (decreased entropy) is expressed as a negative ΔS value whereas decreased order (increased entropy) corresponds to a positive ΔS value.

Each water molecule must participate in hydrogen bonds with other water molecules or with solutes. **Hydrophilic amino acids**, which are on the surface of most proteins, easily form hydrogen bonds with water. This property allows the surrounding water molecules to readily exchange binding partners (ie, they may switch from binding other water to binding hydrophilic amino acids, and vice versa). As a result, the water molecules

surrounding hydrophilic residues are highly disordered. In contrast, the side chains of **hydrophobic amino acids** **cannot form hydrogen bonds** with water. Therefore, water

molecules in the vicinity of hydrophobic residues can only form hydrogen bonds with other water molecules, and they do so by forming a rigid, **highly ordered** network called a solvation layer around the hydrophobic molecule.

Unfolded proteins expose their **hydrophobic residues** to the aqueous environment, causing the surrounding water molecules to form many highly ordered solvation layers around these exposed residues. As proteins fold, these hydrophobic residues aggregate together in the buried core of the protein. This aggregation, also known as **hydrophobic interactions**, minimizes the highly ordered solvation layer and allows hydrophilic residues to remain on the surface of the protein. The minimization of the highly ordered solvation layers allows the water surrounding the folded protein to become **more disordered**. This is reflected by a **positive ΔS** value.

(Choices A and B) The *protein* becomes more ordered upon folding, but the question asks for the order of the *water molecules* surrounding the protein. The water molecules become more *disordered* when the protein folds.

(Choice D) The water molecules become more disordered upon protein folding, but disorder is reflected as a *positive ΔS* rather than a negative ΔS .

Educational objective:

Entropy is a measure of the disorder within a system. Increased order is expressed as a negative entropy change whereas increased disorder corresponds to a positive entropy change. The water molecules surrounding folded proteins have greater entropy than those surrounding unfolded proteins because the hydrophobic molecules on the surface of unfolded proteins force water to form a rigid solvation layer.

Time spent:0 sec

Subject:
Biochemistry

QID:400325

System:
Enzymes